

Newton's Method :-

Newton's method for Quadratic &
Non-Quadratic functions



quadratic function

Converge in one step
for each problem



By Newton method:-

Step 1:- $X_{i+1} = X_i - [J_i]^{-1} \nabla f_i$

Step 2:- Always optimal

Newton's method in one-dimensional

The iterative formula for ~~the~~ finding the roots of the equation $f(x) = 0$ is

$$X_{i+1} = X_i - \frac{f(X_i)}{f'(X_i)}$$

And it is quadratic convergence.

In the next chapter, we will generalize the above study into the Newton's method for n -dimensional problems.

Consider a quadratic function

$$f(x) = \frac{1}{2} x^T A x + B^T x + C$$

Here, A is $n \times n$ symmetric matrix

(named as Hessian matrix).

Expand the function $f(x)$ at $x = x_i$ using the Taylor's Series expansion

$$f(x) = f(x_i) + \nabla f_i^T (x - x_i) + \frac{1}{2} (x - x_i)^T [J_i] (x - x_i)$$

where $[J_i] = [J] |_{x_i}$ is the matrix of second partial derivative (Hessian matrix) order at point x_i .

for minimizing $f(x)$, we have

$$\frac{\partial f}{\partial x} = 0$$

which gives $\nabla f_i + [J_i] (x - x_i) = 0$

If $[J_i]$ is non-singular, then above equation becomes the iterative equation for the method

$$x_{i+1} = x_i - [J_i]^{-1} \nabla f_i$$

Note:- The sequence of the points $x_1, x_2, \dots$ can be shown to converge to the actual solution x^* from any initial point x_1 sufficient close to the solution x^* , provided that $[J_i]$ is non-singular.

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Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$
starting from the point $X_1 = (0, 0)$

The gradient of f is

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

Hessian matrix is

$$J_1 = A = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$\frac{\partial^2 f}{\partial x_1^2} = 1 + 4x_1 + 2x_2 \Rightarrow \frac{\partial^2 f}{\partial x_1^2} = 4$
 $\frac{\partial^2 f}{\partial x_1 \partial x_2} = -1 + 2x_1 + 2x_2, \frac{\partial^2 f}{\partial x_2^2} = 2$
 $\frac{\partial^2 f}{\partial x_2 \partial x_1} = 2, \frac{\partial^2 f}{\partial x_2^2} = 2$

$$\nabla f_1(0, 0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$J_1 \text{ at } (0, 0) = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\text{Thus } J_1^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

Iteration 1:-

$$\text{At } X_1 = (0, 0)^T$$

$$\text{Step 1:- } X_2 = X_1 - [J_1^{-1}] \nabla f_1$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$$

Result:- Newton's method finds the minimum of a quadratic function in one iteration.

Proof:- let the quadratic function be given by

$$f(x) = \frac{1}{2} x^T A x + B^T x + C$$

clearly, $[J_i] = A$

because $J = \nabla^2 f = A$

$$\nabla f = Ax + B$$

By Newton's method

$$x_{i+1} = x_i - [J_i]^{-1} \nabla f_i$$

The minimum of $f(x)$ is given by
 $\nabla f = 0$
 $\Rightarrow x^* = -A^{-1}B$

Step 1:- By Newton's method

$$\begin{aligned} x_{i+1} &= x_i - [J_i]^{-1} \nabla f_i \\ &= x_i - A^{-1} (Ax_i + B) \\ &= x_i - x_i - A^{-1}B \\ &= -A^{-1}B \end{aligned}$$

Hence $x_{i+1} = x^*$ (Hence the result)

Step 2:- check the optimality of x_2

By Newton's method

$$x_{i+1} = x_i - [J_i]^{-1} \nabla f_i$$

Hence, the method has converged in one iteration for the Quadratic function.

Newton's method for non-quadratic function

If $f(x)$ is a non-quadratic function, Newton's method may sometimes diverge and it may converge to saddle points and relative maxima.

This problem can be avoided by modifying the equation with

$$\begin{aligned} x_{i+1} &= x_i + d s_i \\ &= x_i - d_i [J_i]^{-1} \nabla f_i \end{aligned}$$

Where d_i is the minimizing step length in the direction

$$s_i = -[J_i]^{-1} \nabla f_i$$

Newton's method



$J_i = I$ (Identity matrix)

Steepest-Descent
(Cauchy) method

Advantages:-

The modified Newton's method has a number of advantages.

- 1) It will find the minimum in lesser number of steps compared to the original method.
- 2) It finds the minimum point in all cases, whereas the original method may not converge in some cases.
- 3) It usually avoids convergence to a saddle point or a maximum.

With these advantages, this method appears to be the powerful minimization method.

Disadvantages:-

- 1) It requires the storing of a $n \times n$ matrix $[J_i]$.
- 2) It becomes very difficult and sometimes, impossible to compute the elements of the $[J_i]$.
- 3) It requires inversion of the matrix $[J_i]$ at each step.
- 4) It requires the evaluation of the quantity $[J_i]^{-1} \nabla f_i$ at each step.

minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$
 starting from the point $x_1 = (0, 0)$

Sol:-

$$x_2 = x_1 - [J_1^{-1}] \nabla f_1$$

To find x_2

$$[J_1] = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$\boxed{\frac{\partial f}{\partial x_1} = 1 + 4x_1 + 2x_2}, \quad \boxed{\frac{\partial^2 f}{\partial x_1 \partial x_2} = 2}$$

$$\boxed{\frac{\partial^2 f}{\partial x_1^2} = 4}, \quad \boxed{\frac{\partial f}{\partial x_2} = -1 + 2x_1 + 2x_2}$$

$$\boxed{\frac{\partial^2 f}{\partial x_2 \partial x_1} = 2}$$

$$\boxed{\frac{\partial^2 f}{\partial x_2^2} = 2}$$

$$[J_1] = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$[J_1]^{-1} = \frac{1}{(8-4)} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\nabla f_1(0,0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \frac{1}{2} + \frac{1}{2} \\ -\frac{1}{2} - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -3/2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$$

$$\nabla f_2 = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}_{x_2} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \begin{Bmatrix} -1 \\ 3/2 \end{Bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_3 = x_2 - [J_1]^{-1} \nabla f_2$$

$$= \begin{bmatrix} -1 \\ 3/2 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3/2 \end{bmatrix}$$

$\nabla f_2 = 0$, x_2 is the optimum point.

Qo:- $f(x, y) = -x^3 - y^3 + xy + y^2 + xy^2 + 3$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \quad x_2 = x_1 - [J_1]^{-1} \nabla f_1$$

To find x_2 :-

$$[J_1] = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

$$\frac{\partial^2 f}{\partial x^2} = -3x^2 + y + y^2$$

$$\frac{\partial^2 f}{\partial x^2} = -6x, \quad \frac{\partial^2 f}{\partial x \partial y} = 1 + 2y$$

$$\frac{\partial f}{\partial y} = -3y^2 + x + 2y + 2xy$$

$$\frac{\partial^2 f}{\partial y^2} = -6y + 2x$$

$$\frac{\partial^2 f}{\partial y \partial x} = 1 + 2y$$

Q.10: Min $f(x_1, x_2) = x_1^3 + 2(x_1 - x_2)^2 - 3x_1$ starting from point $x_1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ Take 2 iterations?

$$x_2 = x_1 - [J_1]^{-1} \nabla f_1$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 3x_1^2 + 4(x_1 - x_2) - 3 \\ -4(x_1 - x_2) \end{bmatrix}$$

$$x_1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \quad \nabla f_1 = \begin{bmatrix} 3(0.5)^2 + 4(0.5 - 0.5) - 3 \\ -4(0.5 - 0.5) \end{bmatrix}$$

$$= \begin{bmatrix} -2.25 \\ 0 \end{bmatrix}$$

$$J = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}$$

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1 + 4 \Rightarrow \frac{\partial^2 f}{\partial x_1^2} = 6x_1 + 4$$

$$\frac{\partial f}{\partial x_2} = -4(x_1 - x_2), \quad \frac{\partial^2 f}{\partial x_2^2} = 4$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = -4, \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = -4$$

$$J = \begin{bmatrix} 6x_1 + 4 & -4 \\ -4 & 4 \end{bmatrix}$$

$$J_1 \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 7 & -4 \\ -4 & 4 \end{bmatrix}$$

$$[J_1]^{-1} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{7}{12} \end{bmatrix}$$

$$x_2 = x_1 - [J_1]^{-1} \nabla f_1$$

$$= \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{7}{12} \end{bmatrix} \begin{bmatrix} -2.25 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} \frac{1}{3}(-2.25) + \frac{1}{3}(0) \\ \frac{1}{3}(-2.25) + \frac{7}{12}(0) \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} - \begin{bmatrix} -0.75 \\ -0.75 \end{bmatrix}$$

$$= \begin{bmatrix} 1.25 \\ 1.25 \end{bmatrix}$$

$$\nabla f_2 = \begin{bmatrix} 1.6875 \\ 0 \end{bmatrix}$$

$$[J_2]^{-1} = \begin{bmatrix} 11.5 & -4 \\ -4 & 4 \end{bmatrix}$$

$$[J_2]^{-1} = \begin{bmatrix} \frac{2}{15} & \frac{2}{15} \\ \frac{2}{15} & \frac{23}{60} \end{bmatrix}$$

$$= \begin{bmatrix} 1.25 \\ 1.25 \end{bmatrix} - \begin{bmatrix} 0.225 \\ 0.225 \end{bmatrix} = \begin{bmatrix} 1.025 \\ 1.025 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x_3 = x_2 - [J_2]^{-1} \nabla f_2$$

Not optimum

$$= \begin{bmatrix} 1.25 \\ 1.25 \end{bmatrix} - \begin{bmatrix} \frac{2}{15} & \frac{2}{15} \\ \frac{2}{15} & \frac{23}{60} \end{bmatrix} \begin{bmatrix} 1.6875 \\ 0 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.25 \end{bmatrix} - \begin{bmatrix} (\frac{2}{15} \times 1.6875) + \frac{2}{15} x_0 \\ \frac{2}{15} \times 1.6875 + \frac{23}{60} x_0 \end{bmatrix}$$